

Hackathon 2026 - AI-Powered Student Progress & Risk Monitoring

Prototype – Instructional Setup (Running Bash Scripts)

Install

```
python -m venv .venv
```

Windows:

```
.venv\\Scripts\\activate
```

macOS/Linux:

```
source .venv/bin/activate
```

Install dependencies:

```
pip install -r requirements.txt
```

Run the application:

```
streamlit run app.py
```

The terminal local URL:

<http://localhost:8501>

Terminal Local URL: [**http://localhost:8501**](http://localhost:8501)

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Note:

- `http://localhost:8501` will only work after the Streamlit application is actually running on your computer.

Steps:

Check python

- `python --version`

Install Streamlit

- `pip install -r requirements.txt`

Prototype Startup:

- `streamlit run app.py`

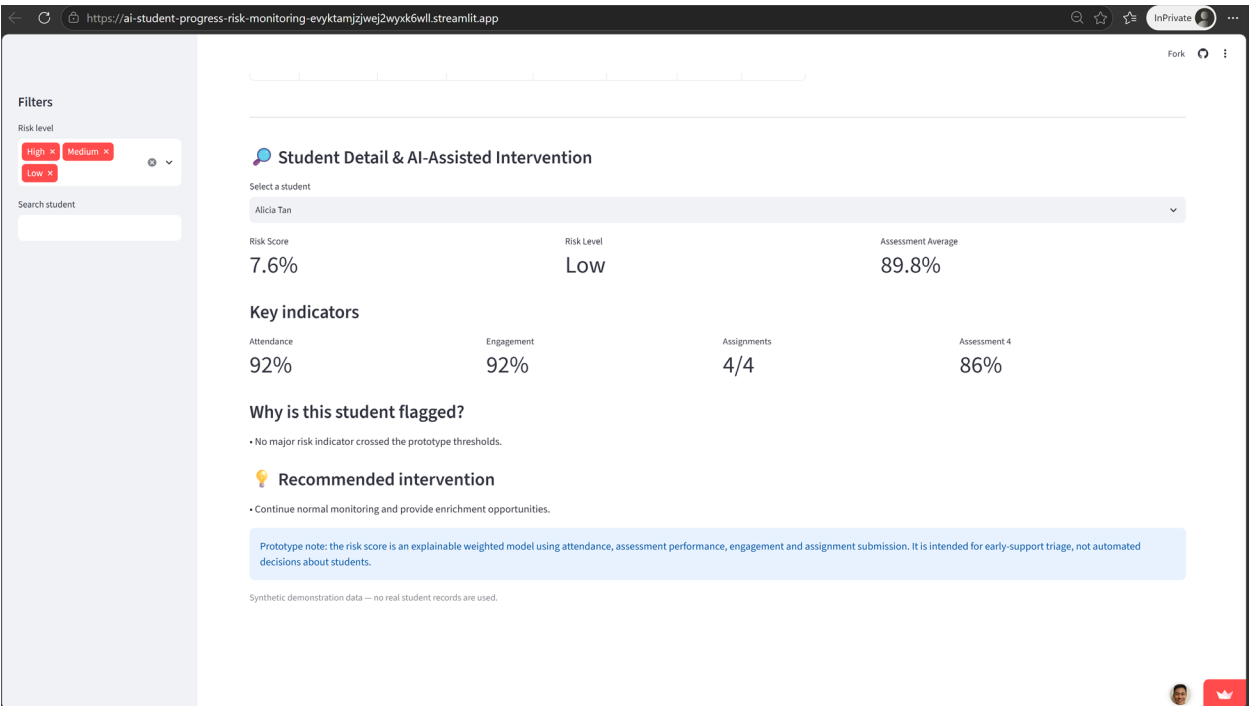
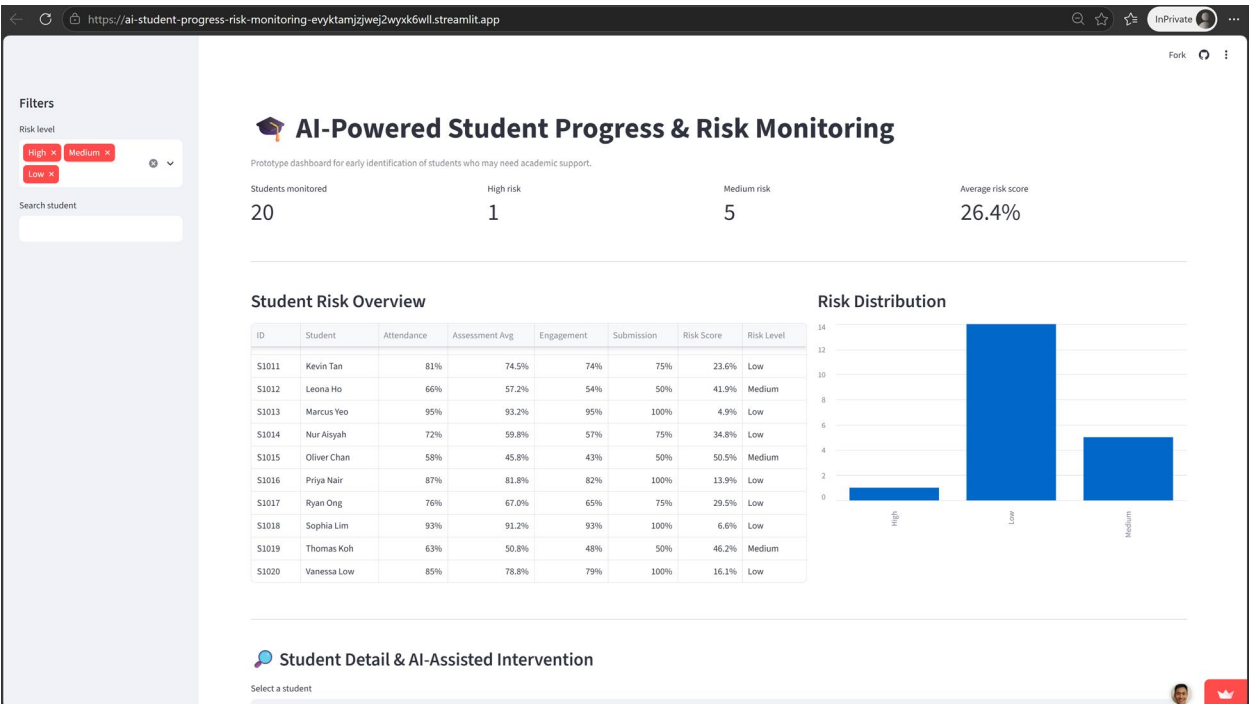
Then follow by:

- <http://localhost:8501>

If streamlit is not recognized, try the following:

- `python -m streamlit run app.py`

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ai-student-progress-risk-monitoring Public

main 1 Branch 0 Tags

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- Prototype.pdf Add files via upload 7 minutes ago
- README.md Add files via upload 4 hours ago
- app.py Add files via upload 4 hours ago
- requirements.txt Add files via upload 4 hours ago

README

AI-Powered Student Progress & Risk Monitoring

1. Solution Summary

This project is a lightweight prototype for helping lecturers identify students who may require early academic support.

About: AI-powered student progress monitoring system that identifies at-risk students using performance, attendance, engagement, and learning activity data.

Releases: No releases published. [Create a new release](#)

Packages: No packages published. [Publish your first package](#)

Contributors: 1

dlimks2002 Derrick Lim

AI_Student_Progress_Risk_Monitoring_Submission > AI_Student_Progress_Risk_Monitoring >			
Sort View ...			
Name	Date modified	Type	Size
Today			
Prototype.pdf	05/10/2026 3:46 PM	Adobe Acrobat D...	167 KB
Prototype.docx	05/10/2026 3:46 PM	Microsoft Word D...	91 KB
README.md	05/10/2026 11:09 AM	Markdown Source...	4 KB
app.py	05/10/2026 10:10 AM	Python.File	5 KB
requirements.txt	05/10/2026 10:10 AM	Text Document	1 KB
docs	05/10/2026 11:07 AM	File folder	
data	05/10/2026 10:10 AM	File folder	

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Directory of files

```
AI_Student_Progress_Risk_Monitoring
|
├─ app.py
├─ requirements.txt
├─ README.md
├─ data
│   └─ students.csv
└─ docs
```

```
AI_Student_Progress_Risk_Monitoring/
├─ app.py                ← Working Streamlit prototype
├─ requirements.txt       ← Python dependencies
├─ README.md             ← Documentation & setup
├─
├─ data/
│   └─ students.csv      ← 20 synthetic students
├─
├─ docs/
│   └─ Project_Documentation.pptx ← Submission PowerPoint
│   └─ 3_Minute_Demo_Script.md    ← Video narration script
├─
└─ assets/
```

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Workflow Demonstration

1. Open the dashboard.
2. Show the overall student monitoring KPIs.
3. Filter to **High Risk** students.
4. Select Learners such as **Isaac Goh** or **Farah Ahmad**.
5. Show the risk score and indicators.
6. Explain why the student is flagged.
7. Show the recommended intervention.
8. Return to the overview and explain how a lecturer could use the dashboard for early intervention.

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Project Link

Repository placeholder:

<https://github.com/dlimks2002/ai-student-progress-risk-monitoring>

Prototyping model could be further enhanced with the following:

Future Enhancements

- Train a supervised ML model using historical anonymised student data.
- Add weekly progress trend charts.
- Integrate LMS/student information system data through APIs.
- Add lecturer notes and intervention tracking.
- Add model evaluation metrics and fairness monitoring.
- Add role-based access control and privacy safeguards.
- Use an LLM/RAG layer to generate personalised learning-resource recommendations.

AI Concern:

The prototype should support lecturers rather than replace professional judgement. Risk flags can be affected by incomplete data, contextual factors or bias. Any real deployment should use appropriate governance, access control, data minimisation, transparency, human review and model monitoring.

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What the prototype does for us

The dashboard includes:

- **20 leaners**
- Attendance monitoring
- Assessment performance
- Engagement percentage
- Assignment submission
- Explainable **risk score**
- **Low / Medium / High** risk classification
- Risk distribution chart
- Student search/filter
- Individual student drill-down
- Explanation of why a student is flagged
- Suggested lecturer interventions
- Responsible-AI disclaimer

The risk model is deliberately simple and explainable:

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Risk Score =

- 35% assessment performance
- 30% attendance
- 20% engagement
- 15% assignment submission

This is a good choice for a **prototype demonstration** because we can clearly explain how the system works instead of claiming that a complex AI model magically predicts student failure.

Running the Project

After doing the ZIP Extraction - Running it:

```
cd AI_Student_Progress_Risk_Monitoring
```

```
python -m venv .venv
```

```
.venv\Scripts\activate
```

```
pip install -r requirements.txt
```

```
streamlit run app.py
```

Streamlit URL shown:

```
http://localhost:8501
```

Recommendation Data Testing

- We would recommend using **Isaac Goh** as the main high-risk example for the demonstration.

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Submission Prototype URL

Live Prototype:

<https://ai-student-progress-risk-monitoring-evyktamjzjwej2wyxk6wll.streamlit.app/>

Source Code:

<https://github.com/dlimks2002/ai-student-progress-risk-monitoring>